UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

Project Report

Course Title: Computer Networks

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1. Project Title

GRANDSTAND CONNECT: A VLAN-Based Smart Stadium Network

2. Abstract

This project presents the design and implementation of a secure, scalable, and logically segmented smart stadium network using Virtual Local Area Networks (VLANs). The proposed network supports multiple operational zones including Administration, Ticketing, Guest Access, Security, and Media services, each isolated into separate VLANs to enhance security, performance, and manageability.

The network was implemented using Cisco Packet Tracer. Inter-VLAN communication was achieved using the Router-on-a-Stick architecture, while Dynamic Host Configuration Protocol (DHCP) was configured to automate IP address assignment across all zones. Dedicated servers were deployed to simulate real-world services such as administrative management, ticketing systems, and media/scoreboard services. The project demonstrates how structured network design improves efficiency, reduces broadcast traffic, and enables secure service delivery in large public venues like stadiums.

3. Introduction

Modern stadiums rely heavily on robust network infrastructures to support administrative operations, ticketing systems, public internet access, security monitoring, and live media broadcasting. Without proper segmentation and centralized control, such environments suffer from performance degradation, security vulnerabilities, and complex management challenges.

The objective of the *Grandstand Connect* project is to simulate a real-world enterprise-level smart stadium network using VLANs, DHCP, and inter-VLAN routing techniques. This project provides practical exposure to modern network design principles while ensuring logical separation between departments with controlled connectivity through a centralized core network.

4. Problem Definition

Large venues such as stadiums host a diverse range of users and devices operating simultaneously. Traditional flat network architectures lead to excessive broadcast traffic, poor security, and inefficient resource utilization.

This project addresses the problem of network congestion and security risks caused by the absence of logical segmentation. The scope of the project is limited to LAN-level design and simulation. Assumptions include fixed device counts per zone and wired connectivity. WAN integration and real-world hardware constraints are beyond the scope of this implementation.

5. Literature Review

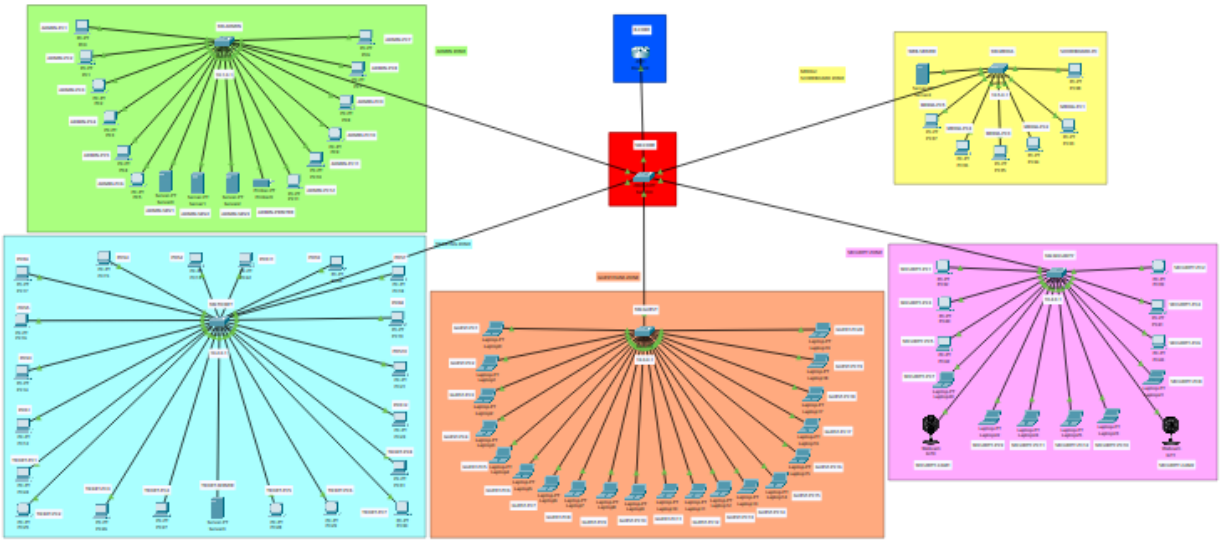
Existing enterprise network designs widely adopt VLAN-based segmentation and hierarchical architectures to manage large-scale environments. Research and industry practices indicate that VLANs significantly improve network performance and security by isolating departmental traffic. DHCP is commonly used to reduce manual configuration errors, while Router-on-a-Stick provides a cost-effective solution for inter-VLAN routing in medium-scale networks.

However, many academic implementations remain theoretical and lack realistic service simulation. This project enhances realism by integrating functional web-based services mapped to different network zones.

6. System Architecture

The network follows a hierarchical architecture consisting of a core layer, access layer, and routing layer. A central core switch manages VLAN trunking, while individual access switches serve each operational zone. A single router performs inter-VLAN routing using sub-interfaces. End devices include PCs, servers, and monitoring systems.

Each operational zone is assigned a dedicated VLAN to ensure logical isolation and controlled communication.



7. Network Design

A structured IP addressing scheme was implemented by assigning a dedicated subnet to each VLAN, ensuring logical segmentation, efficient traffic management, and enhanced security. Each zone operates within its own IP range with a unique default gateway configured on the

router using Router-on-a-Stick. This design simplifies network management, reduces broadcast traffic, and enables controlled inter-VLAN communication.

VLAN	Zone	Network	Gateway
10	Admin	10.1.0.0/24	10.1.0.1
20	Ticketing	10.2.0.0/24	10.2.0.1
30	Guest	10.3.0.0/24	10.3.0.1
40	Security	10.4.0.0/24	10.4.0.1
50	Media	10.5.0.0/24	10.5.0.1

VLAN configuration and Trunk link configuration on access switch:

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   ADMIN                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
20   TICKETING                        active
30   GUEST                            active
40   SECURITY                         active
50   MEDIA                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/17      auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Fa0/17      1-1005

Port        Vlans allowed and active in management domain
Fa0/17      1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/17      1,10,20,30,40,50

Switch#
```

Trunk configuration on core switch:

```
Switch>enable
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50
Fa0/3       10,20,30,40,50
Fa0/4       10,20,30,40,50
Fa0/5       10,20,30,40,50
Gig0/1      10,20,30,40,50

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50
Fa0/3       10,20,30,40,50
Fa0/4       10,20,30,40,50
Fa0/5       10,20,30,40,50
 --More-- |
```

8. Protocols and Technologies Used

The following protocols and tools were used in this project:

- **VLAN (IEEE 802.1Q)** for logical segmentation
- **DHCP** for automated IP address allocation
- **TCP/IP** for end-to-end communication
- **HTTP** for web-based service simulation
- **Cisco Packet Tracer** for network simulation

These technologies were selected due to their industry relevance and suitability for enterprise networks.

9. Implementation Details

VLANs were configured on all switches, and trunk links were established between access switches and the core switch. Router sub-interfaces were created to enable inter-VLAN routing. DHCP pools were configured per VLAN to dynamically assign IP addresses to end devices. Servers were assigned static IP addresses to ensure service consistency.

Different servers hosted web services representing administrative management, ticketing operations, and media/scoreboard display to simulate real-world usage.

Router sub-interfaces for inter-VLAN routing:

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.1.0.1        YES manual up                    up
GigabitEthernet0/0.20  10.2.0.1        YES manual up                    up
GigabitEthernet0/0.30  10.3.0.1        YES manual up                    up
GigabitEthernet0/0.40  10.4.0.1        YES manual up                    up
GigabitEthernet0/0.50  10.5.0.1        YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

DHCP address allocation for VLANs:

```
Router>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.1.0.50        000B.BE74.A581           --                     Automatic
10.1.0.51        00E0.A339.AA45           --                     Automatic
10.1.0.52        0001.96DA.814E           --                     Automatic
10.1.0.53        0040.0BE4.792E           --                     Automatic
10.1.0.57        0090.2187.5C10           --                     Automatic
10.1.0.58        0060.2F22.D877           --                     Automatic
10.1.0.54        000D.BD63.46AB           --                     Automatic
10.1.0.55        0030.F272.34B1           --                     Automatic
10.1.0.59        0009.7CD4.B2A2           --                     Automatic
10.1.0.56        0004.9AC6.CA3C           --                     Automatic
10.1.0.60        0060.5C55.5696           --                     Automatic
10.1.0.61        00E0.B08B.6BED           --                     Automatic
10.1.0.62        00D0.5829.A88E           --                     Automatic
10.2.0.58        00E0.F781.B94D           --                     Automatic
10.2.0.51        000C.8554.7ED1           --                     Automatic
10.2.0.59        0002.4AA7.D063           --                     Automatic
10.2.0.53        0001.6480.2036           --                     Automatic
10.2.0.55        0001.C764.5A13           --                     Automatic
10.2.0.52        00E0.A3A6.2744           --                     Automatic
10.2.0.50        000A.41B2.CC24           --                     Automatic
 --More-- |
```

10. Performance Evaluation

The network was evaluated based on successful DHCP operation, inter-VLAN connectivity, and service accessibility across all configured zones. Ping tests confirmed reliable communication between authorized VLANs, while DHCP binding tables verified correct and consistent IP address allocation to all end devices.

The network demonstrated stable performance with near-zero packet loss and minimal latency within the LAN environment. Efficient VLAN segmentation and optimized routing ensured smooth traffic flow, reduced broadcast overhead, and consistent service availability under normal operational conditions.

11. Security Considerations

- VLAN-based segmentation was implemented to isolate different departments and restrict unauthorized access.
- Guest users were confined to a dedicated VLAN, preventing access to sensitive administrative and ticketing resources.
- Centralized inter-VLAN routing was used to control communication paths between network segments.
- Logical separation of services minimized internal security risks and reduced the overall attack surface.
- Controlled IP allocation through DHCP improved device traceability and network monitoring.

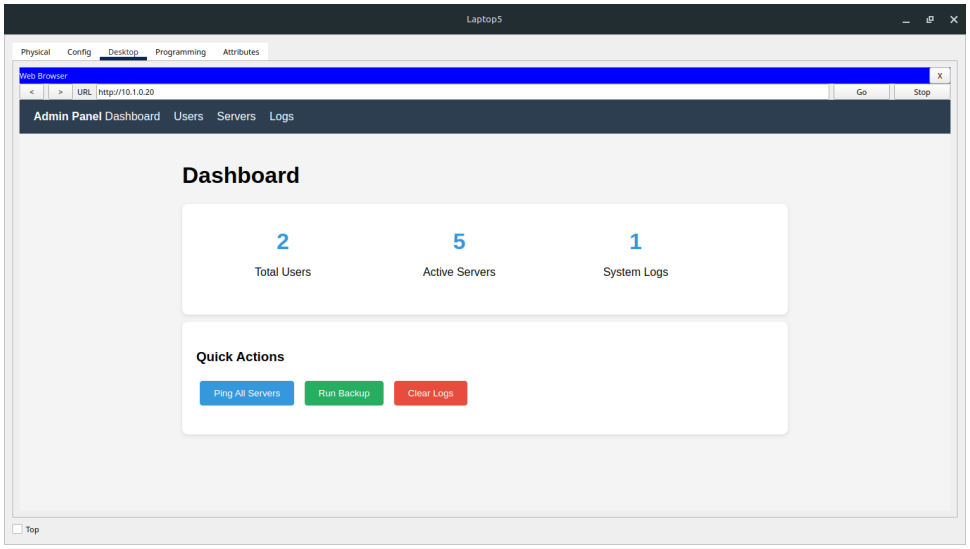
12. Results and Analysis

The implemented network successfully met all defined design objectives. All devices received correct IP addresses through DHCP, VLAN segmentation operated as intended, and inter-VLAN communication functioned reliably. All deployed services, including administrative access, ticketing operations, and media display systems, performed as expected.

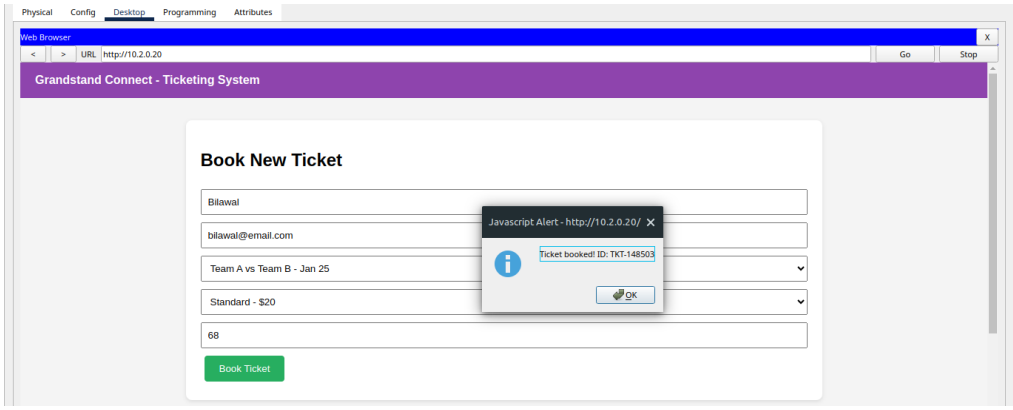
The inclusion of functional web-based services added realism to the simulation and demonstrated the practical applicability of the design. The results validated the efficiency, scalability, and reliability of the proposed smart stadium network architecture.

Web-based service accessed from client PC(Simulation only)

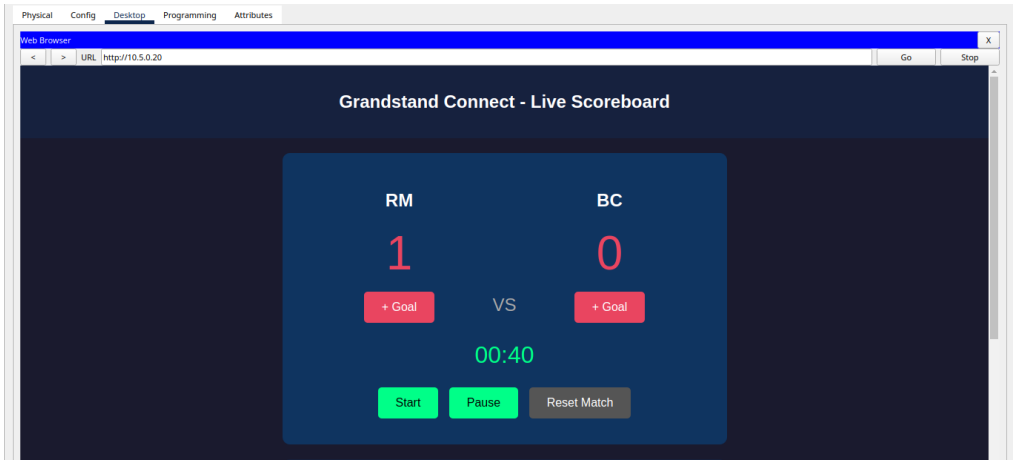
Admin can add user's with their role, check servers and contol system logs:



Guests can book tickets:



Fans can watch live score:



13. Discussion

The project successfully demonstrates core enterprise networking principles such as VLAN-based segmentation, inter-VLAN routing, and centralized IP management using DHCP. Although implemented in a simulated environment, the design closely resembles real-world enterprise deployments, providing strong conceptual clarity and practical networking experience.

The integration of web-based services further enhanced the realism of the system, demonstrating how networking and application layers work together in a smart infrastructure environment.

14. Challenges and Limitations

Several challenges were encountered during the implementation, including limited configuration capabilities for IoT devices and security appliances in Cisco Packet Tracer. Additionally, the absence of real hardware restricted advanced troubleshooting and performance benchmarking.

The project is also limited to wired LAN implementation and does not include wireless networking, advanced firewall configurations, or external WAN connectivity.

15. Applications and Use Cases

The proposed network design can be applied to:

- Smart stadiums and sports complexes
- University and college campuses
- Corporate office environments
- Public event venues and convention centers
- Large-scale institutional LAN infrastructures

16. Conclusion

The *Grandstand Connect* project successfully implements a secure, scalable, and efficient smart stadium network using VLAN segmentation, DHCP-based IP management, and Router-on-a-Stick inter-VLAN routing. The objectives of performance optimization, security enhancement, and service integration were fully achieved.
Overall, the project provides a strong foundation in enterprise networking concepts and demonstrates practical skills required for real-world network design and deployment.

17. Future Enhancements

Future improvements may include:

- Wireless access point integration for mobile connectivity
- Advanced ACLs and firewall-based security policies
- Cloud-based hosting of web services
- Network monitoring, logging, and traffic analysis systems
- Redundant routing and failover mechanisms for high availability

18. References

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